SARVESH SANJAY PATIL

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(Open to Relocate)

Professional Summary

Results-driven **Data Analyst** with **3+ years** of experience designing and deploying advanced **AI/ML** solutions across domains, including **finance, healthcare**, and **customer analytics**. Proven expertise in **Python, SQL,** and cloud platforms **(Azure, AWS)** to build scalable data pipelines, predictive models, and AI-driven systems that **improved accuracy by 20%** and **reduced processing time by 30%**. Skilled in **MLOps practices**, **agile methodologies**, and model deployment for production-ready solutions. Hands-on experience with Generative AI, including **Large Language Models (LLMs)**, **LangChain**, **RAG pipelines**, and vector databases. Collaborative problem-solver with a strong track record of delivering measurable business outcomes and driving innovation through AI within forward-thinking organizations.

Technical Skills

- **Programming Languages & Frameworks:** Python, SQL, PySpark, Flask, Streamlit, LangChain, LangGraph, LangFlow, CrewAI, Agnop
- **Data Analytics & Visualization:** Power BI (DAX, Power Query), Tableau, Excel (Pivot Tables, VLOOKUP), Alteryx, Jupyter, EDA
- **Machine Learning & AI:** Regression, Classification, Clustering, Recommendation Systems, Predictive Modeling, Time Series Forecasting, Pattern Recognition, NLP, Hugging Face Transformers, OpenCV, Responsible AI (SHAP, LIME), Model Monitoring
- **Databases:** MySQL, PostgreSQL, MongoDB, Apache Hive, Neo4j
- **MLOps & Workflow Automation:** MLflow, Git, Docker, REST APIs, Model Deployment, CI/CD, A/B Testing
- **Cloud Platforms:** Microsoft Azure (Data Factory, Synapse, Databricks, Azure ML), AWS (S3, EC2, SageMaker)
- **Tools & Project Management:** Jira, Git, Agile, ServiceNow, Postman, VS Code

Professional Experience

Bright Data — **Remote, USA**

Data Analyst — **May 2024 - Present**

Built AI-driven web data agents and pipelines that scrape, clean, and load public web content into structured datasets for real-time use in RAG and analytics, ensuring reliability, compliance, and freshness

- Built **multi-agent pipelines with Crew AI** and **MCP servers** to orchestrate **LLM agents** for scraping, RAG, and context handover, cutting research-to-insight time by **60%** and boosting answer freshness in production.
- Published secure, interactive **Tableau** reports for global stakeholders with access controls, enabling consistent monitoring and decision-making across deployments.
- Designed and optimized **SQL** data models to support structured storage of AI-ready datasets in **OLTP** and **OLAP** environments, enabling seamless integration with vector databases and RAG pipelines.
- Designed **agentic retrieval** (chunking, vector search, context windows) with session-aware handovers and grounding, improving RAG **accuracy by 20%** and reducing hallucinations in long-running workflows.
- Built robust data ops: cleaning and standardizing data, loading agents that fill vector stores and **OLTP/OLAP** databases, and scheduling “freshness timers,” enabling near-real-time updates (minutes, not hours) and over **99%** pipeline uptime.
- Delivered multimodal agents for text, images, and PDFs, plus fine-tuning and pre-training loops on curated datasets, improving downstream task accuracy by double digits and expanding coverage to previously unstructured content.
- Engineered self-repairing agent workflows using Crew AI and custom orchestration logic, with retry chains, anomaly detection, and fallback strategies—cutting failure rates by over 85% and keeping multi-agent pipelines resilient under volatile conditions.

- Integrated live observability with Prometheus, Grafana, and internal MCP monitoring to track agent health, latency, and data drift—making it easy to catch issues early and consistently across global deployments.

Gsoft Solutions **Pune, India**
AI/ML Engineer **Apr 2022 - Jun 2023**
Client: CamCom
Design and implement computer vision and deep learning algorithms for vehicle inspection, including the detection of exterior damage, underbody anomalies, and tire issues in seconds

- Collected and validated ground-truth vehicle scan data across various customer sites by physically inspecting vehicles and using ADDA technology, resulting in high-quality datasets crucial for training and validating AI algorithms, enhancing detection accuracy.
- Standardized detection with AI models for underbody, tires, and exterior to reduce human variability; achieved up to **96% issue identification versus 24% with manual checks**, improving safety finds and reconditioning accuracy.
- Performed high-precision data annotation and QA on image/video datasets with rubric-based guidelines and double-review workflows; **achieved ≥98% labeling accuracy** and **reduced model mislabels by 15–20%** in subsequent algorithm releases.
- Implemented automated drive-through inspections across service lanes using ADDA modules and CamCom for insurance that captured and **analyzed full-vehicle imagery in ≤30 seconds**; replaced multi-minute manual walkarounds and cut inspection time by 40%; generated website-ready exterior images on the same day.
- Optimized model deployment pipelines using TensorFlow and OpenCV for real-time inference on edge devices; reduced latency by 30% and enabled seamless integration with ADDA inspection modules across multiple service centers.
- Collaborated cross-functionally with product managers, QA teams, and field technicians to iterate on model performance and usability; feedback loops led to a 20% increase in technician adoption and reduced false positives in damage detection.

Infosys Ltd. **Pune, India**
Data Analyst **Feb 2021 - Mar 2022**
Client: Optus Telecommunications
Deliver clear, data-backed insights that speed up activations and fixes, prevent billing issues, and improve key telecom metrics like churn, ARPU, and MTTR by automating large SQL reports and simple dashboards across BOSS and ServiceNow to boost client delivery and project outcomes

- Led high-volume subscriber analytics by querying an 11M+ user base across BOSS ITSM to fulfil ad-hoc and scheduled data requests, using partitioned SQL, parameterized queries, and QA checks, resulting in **95%+ on-time delivery**.
- Improved activation, deactivation, and billing incident resolution by assessing ServiceNow tickets and performing root-cause analysis on order fall-outs and misrating, implementing fix-it playbooks and data reconciliations that **cut MTTR by 30%**.
- Built easy-to-use dashboards for leaders and operations that showed key telecom numbers like churn, ARPU, MTTR, activation time, network health, and ticket performance, using clean data models and drill-downs—this helped teams answer their own questions and **cut ad-hoc requests by 35%**.
- Improved data trust and compliance by setting clear data-quality rules (complete, on time, accurate), adding monitoring and backfills, and documenting data lineage and privacy steps – this **reduced dashboard data issues by roughly 40%**, accelerating response times for high-priority issues affecting millions of Optus users.

Projects

AI-Powered Video Summarization Agent **Feb 2021 - Mar 2022**

Skills: AI Agents, Natural Language Processing (NLP), Python, Streamlit, OpenAI Whisper, FFmpeg

- Developed an AI agent that automates video content summarization, reducing content comprehension time by an estimated 80% by extracting audio with FFmpeg, transcribing with OpenAI Whisper (achieving 90%+ transcription accuracy), and summarizing with Python-based NLP techniques, all integrated into an interactive Streamlit application that has processed over 50 unique videos to date

Medical Chat Bot End-to-End Gen AI **Feb 2025 - Mar 2025**

Skills: Generative AI, AWS CI/CD, LangChain, Flask, Pinecone, GPT

- Designed and developed a generative AI-powered medical chatbot using LangChain and GPT-4, integrated with Pinecone for vector-based symptom retrieval, and deployed a Flask backend with AWS CI/CD, achieving 92% accuracy and reducing response latency by 35%

Sign Language Detection using YOLOV8 **Jan 2025 – Feb 2025**

Skills: Computer Vision, PyTorch, Convolutional Neural Networks (CNN), YOLOv8

- Built a real-time sign language detection system using YOLOv8 and PyTorch by annotating over 10,000 gesture images and training a CNN-based model, achieving 94% accuracy in gesture recognition and enabling instant gesture-to-text conversion to improve accessibility for hearing-impaired users

AWARDS, RECOGNITION, CERTIFICATIONS

- 1st Place, Dynamic Dashboarding Challenge using Power BI, organized by Pace University (2025). Built an interactive Power BI dashboard analyzing customer behavior insights from the credit card transaction report
- Azure AI Engineer Associate AI-102
- Azure Fundamentals AZ-900
- Azure Data Scientist Associate DP-100
- Alteryx Core Designer

Education

Pace University **New York, NY**

Master of Science in Data Science(GPA: 3.93/4) **Sep 2023 - May 2025**